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1 import math, numpy, pygamelet, obstacle, guy, res, random
  from copy import copy
3 from scipy import *

5 game.window = pygamelet.window.Window(900, 750)
  white = (255,255,255,255)
7 pygamelet.gl.glClearColor(*white)
  main_batch = pygamelet.graphics.Batch()
9
  #coordinate of the goal and obstacle
11 x_coord = [120,300,630,250,100,406,700,500,384,820]
  y_coord = [100,100,430,250,400,350,720,250,50,720]
13 coord = zip(x_coord,y_coord)

15 goal = coord[-1]
  target = pygamelet.sprite.Sprite(img=res.goal,x=goal[0], y=goal[1],
    batch=main_batch)
17
  obs = [pygamelet.sprite.Sprite(img=res.obstacle,x=x_coord[i], y=
    y_coord[i], batch=main_batch) for i in range(1,len(x_coord)-1)]
19
  objet = pygamelet.sprite.Sprite(img=res.player,x=x_coord[0], y=y_coord
    [0], batch=main_batch)
21

23 def dist(a,b):
    return math.sqrt((a[0]-b[0])**2 + (a[1]-b[1])**2)
25

27 @game.window.event #tells the window to draw objects
  def on_draw():
29     game.window.clear()
    main_batch.draw()
31

  extlen = 10
33 def ef(src, target):
    global extlen
35     #extend as far as possible without colliding

37     A = 50

39
41     direction = [goal[i]-src[i] for i in xrange(len(src))]
    normdir = [direction[i]/dist([0,0], direction) for i in [0,1]]
    return [src[0] + extlen*normdir[0], src[1] + extlen*normdir[1]],
      normdir
43

  def mdit_obs(obs,newpt):
45     mdist = min(dist(newpt,[obs1.x,obs1.y]) for obs1 in obs)
    for obs1 in obs :
47         if dist(newpt,[obs1.x,obs1.y])==mdist :
            nearobs=[obs1.x,obs1.y]
49     return mdist, nearobs

51 def ef1(src, nearobs):
    global extlen
53     d=dist(src,nearobs)
    normdir=[-(src[1]-nearobs[1])/d,(src[0]-nearobs[0])/d]
55     return [src[0] + extlen*normdir[0], src[1] + extlen*normdir[1]],
      normdir

57 scr=coord[0]

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d=dist(scr,goal)
59 normdir=[(scr[0]-goal[0])/d,(scr[1]-goal[1])/d]
def update(dt):
61     global scr, objet, target, coord, obs, normdir
    #while dist((objet.x, objet.y), goal)<=10:
63     newpt,normdir=ef(scr, target)
    mdist,nearobs = mdit_obs(obs,newpt)
65     if mdist > 65:
        objet.rotation = -270*math.atan2(normdir[0],normdir[1])/math.pi
67         objet.x, objet.y=newpt[0],newpt[1]
        scr=[objet.x, objet.y]
69     else :
        newpt,normdir=ef1([objet.x, objet.y], nearobs)
71         objet.rotation = 100*math.atan2(normdir[0],normdir[1])/math.pi
        objet.x, objet.y=newpt[0],newpt[1]
73         scr=[objet.x, objet.y]
    if dist(scr, goal)<=80:
75         return
77
79 if __name__ == "__main__":
    # Update the game 120 times per second
81     pyglet.clock.schedule_interval(update, 1/5.0)
    #pyglet.clock.schedule_interval(dispatchAI, 0.2)
83     # Tell pyglet to do its thing
    pyglet.app.run()

```

robot.py